Security Attacks - Threats - Risks - Security Goals

Security Attacks	Threats	Risks	Security Goals
Phishing Tick users to reveal Sensitive information (eg. login details).	Social engineering threat (users disclose confidential data)	Loss of confidentiality (exposure of sensitive information)	**Confidentiality** Protects data from unauthorized access (e.g. access, MFA)
Malware Malicious software that injects or damages	Malware threat (System gets infected/ data altered)	Data corruption or loss (files encrypted / deleted)	**Integrity** Ensures data is accurate and unaltered (e.g digital)
DoS / DDoS Overloads a System with heavy traffic, making it unavailable.	Service disruption threat (System/Service becomes unavailable)	Service unavailability (downtime / loss of Service)	**Availability** Ensures Systems and Services are accessible (e.g) firewalls, DDoS Protection, backups).
Unauthoried Access Gaining access without permission (e.g, using stolen credentials).	Data theft threat (unauthorised access to data / system)	Financial loss/ reputation damage (fraud, data leakage loss of trust)	Confidentiality + Integrity Prevents unauthorized access and malicious changes (e.g Strong intrusion detection)
Data Breach Exposing sensitive data due to vulnerabilities or insider threats.	Confidentiality breach threat (Sensitive data exposed)	Data leakage / regulatory Penalties (Violation of data protection laws)	Confidentiality / Integrity + Availability Comprehensive protection (eg. encryption)

Substitution Techniques - Short Summary

Caesar Cipher (Shift Cipher)

Key:
Shift 0-25

Complexity
very low

Security:
very weak

Example:
HELLO, Key 3 → KHOOR

Monoalphabetic Cipher (Fixed Substitution)

Key:
fixed substitution alphabet

Complexity:
Low - moderate

Security:
weak - moderate

Letters mapped consistently

Vigenere Cipher (Polyalphabetic Substitution)

Key:
repeated Keyword

complexity:
Moderate - high

Security
Stronger than Caesar

Example:
HELLO, key KEY →
RIJVS

Hill Cipher (Matrix - based Substitution)

Key:
invertible matrix

Complexity:
High

Security:
Strong among these four

Uses matrix multiplication mod 26

Security generally increases: Caesar → Monoalphabetic → Vigenere →
→ Hill

Classical Encryption Schemes - Substitution vs Transposition Techniques (Short Summary)

Substitution Techniques
(Replace characters with others (chars))

Caesar Cipher
* Shift letters by fixed number (Key – 0-25)

Ex:
HELLO (Key 3)
→ KHOOR

Monoalphabetic cipher
* Fixed substiti mapping for each letter.

Ex:
HELLO (Key: QWERTY)
→ CRJKKW

Vigenere cliphar
Uses a keyword (repeated) for Substiution.

Ex:
HELLO (Key: LEMON
→ 70YHF

Hill ciphar.
Uses matrix multiplication on blocks of letters.

Ex:
HELLO (2×2 mat
→ 72QWE

Transposition Techniques.

Rail Fence Cipher
* Writes text in a zigzag pattern over rails and reads row wise

Example:
HELLO (3 rails)
→ HOLEL

Columnar Transposition Cipher
Arranges text in matrix using a key

example:
HELLO (Key: ZEBRA)
→ ELHLO

Similarities
* Both are clasical encryption methods
* Both use a key for encryption and decryption
* Both are reversible (decryption restores original text).
* Both operate on plaintext characters
* Both provide confidentiality by transforming the data

Comparison of Subsitiution and Transposition Techniques.

Feature	Caesar	Monoalphabetic	Vigenere	Hill	Rail Fence	Columnar Transposition
Main Idea	Shift letters	Fixed letter mapping	Keyword	Matrix multipitication	Zigzag arrangement	Ready by columns (Key)
Data Structure	Same Structure	(only values change)	Same structure	Blocks are changed	order of Characters Changes	order of Characters Changes
Effect	(only values Change)	Low to moderate (Vulnerable to frequency analysis)	(Values changes using key)	High (resists Simple frequency)	Low to moderate	Moderate (Vulnerable to pattern analysis especially Short Key).
Security Level	very low		Moderate to high			

STEGANOGRAPHY

Hiding information in plain sight

Data HIDING

* Hides secret data inside an images, audio, video, or text
* Focuses on Secrecy.

WATERMARKING

* Adds a mark or ID to Protect ownership.
* Helps detect tampering

SURVIVABILITY

* Keeps hidden data safe after compression, resizing, or attacks.
* Ensure reliability.

HOW THEY WORK TOGETHER

* Data hiding embeds the message
* watermarking proves ownerships
* Survivability keeps data intact
* Together they improves secure communication.

SECURITY BENEFITS

DATA HIDING
Secrecy and privacy

WATERMARKING
ownership and authenticity

SURVIVABILITY
Robustness and reliability

APPLICATIONS

Secret message in an images (LSB)

Copyright logo in digital media.

Watermarks that survive compression.